

# Veera Manikanta Gonugondla

Incoming Operations Analyst, Goldman Sachs, Hyderabad

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## SUMMARY

Electrical & Electronics Engineering student who designs digital hardware **from the instruction set upward** and gets it running on a board. Current work is MAK8u, a **dual-core microcontroller in SystemVerilog**, closed post-route on an Artix-7 at **50 and 100 MHz** with **126 testbenches** behind it. Also wrote RTL through commercial EDA flows to synthesis during an internship, and brought up SPI, I2C and RS-485 with an ILA and a scope. Co-inventor on **two Indian patent applications; Systems & Digital Engineering Intern at Boeing India**.

## TECHNICAL SKILLS

|                |                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages      | SystemVerilog, Verilog, C, Embedded C, Assembly, Python, MATLAB, Tcl, Bash                                                                                                                                      |
| Digital Design | RTL design, computer architecture, pipelined in-order cores, hazard detection and forwarding, FSM and datapath design, clock-domain crossing, memory arbitration, interrupt and DMA controllers, ISA definition |
| Verification   | Self-checking SystemVerilog testbenches, automated mutation testing, functional coverage, waveform and ILA debugging, synthesis and post-route timing closure                                                   |
| Tools          | AMD Vivado, Verilator, Nexys A7 (Artix-7), Git, Linux, Simulink, LTspice, LabVIEW, oscilloscopes and logic analyzers                                                                                            |
| Embedded       | Raspberry Pi, ESP32, Arduino, FreeRTOS, MQTT, MODBUS RTU / RS-485, PCB-level debugging                                                                                                                          |

## EDUCATION

### BTech, Electrical & Electronics Engineering

Jul 2023 – Sep 2027

Manipal Institute of Technology, Manipal · **CGPA 8.07 / 10** through 6th semester

Minor Computational Intelligence

### BS, Electronic Systems

Jul 2023 – Dec 2027

Indian Institute of Technology Madras · **CGPA 7.33 / 10** through 7th semester

Program Open-entry online BS, taken concurrently with the on-campus BTech at Manipal

## CERTIFICATIONS

Computer Architecture Essentials on Arm — Arm Education (Aug 2026)

FPGA Architecture Based System for Industrial Applications — L&T EduTech (Aug 2026)

## PROJECTS

### MAK8u — Asymmetric Dual-Core Microcontroller

Jun 2025 – Present

|              |                                                                                                                                                                                                                                                                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview     | A dual-core 8-bit microcontroller built from its own instruction set up. One core runs <b>five stages at 50 MHz</b> for predictable timing, the other <b>seven at 100 MHz</b> for throughput, and one MMCM holds both at an <b>exact 2:1 ratio</b> , so the boundary between them closes under static timing analysis rather than through a synchronizer    |
| Feature set  | Forwarding and hazard logic on both cores, an inter-core controller with message queues, semaphores and barriers, a lock arbiter that <b>reports the worst-case wait</b> before a core commits to it, DMA, and SPI, I2C, UART, GPIO, PWM and XADC peripherals                                                                                               |
| Result       | Both clocks close post-route, <b>WNS +0.452 ns and +0.480 ns</b> with positive hold slack, in 8,823 LUTs and 7,619 registers on an Artix-7. Crossing latency held at <b>2 cycles across 27.5M silicon measurements</b> , versus 32 to 36 with an asynchronous crossing. Shortened the critical path to the program counter <b>from 14 logic levels to 6</b> |
| Verification | <b>126 self-checking testbenches</b> , all passing before every synthesis. Five <b>mutation campaigns</b> found the tests that only looked strong                                                                                                                                                                                                           |
| Tooling      | A <b>54-instruction ISA</b> with a Python assembler, ILA capture for silicon debug, and MAK IDE, a terminal workbench that loads images over UART without Vivado                                                                                                                                                                                            |
| Achievements | <b>Patent draft in progress</b> on repeatable inter-core timing across rationally related clock domains                                                                                                                                                                                                                                                     |
| Stack        | SystemVerilog, Vivado, Tcl, Verilator, Python, Nexys A7 (Artix-7)                                                                                                                                                                                                                                                                                           |

## PROJECTS (CONT.)

### Project Garuda — Edge-AI Home Security & Surveillance

Jan 2025 – Present

|              |                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Overview     | A surveillance stack that runs detection on the device itself. No cloud, and no video leaves the home                                     |
| Stack        | Raspberry Pi 5, Hailo-8L M.2 AI HAT (13 TOPS), YOLOv8s, Python                                                                            |
| Result       | INT8-quantized inference at <b>52.2 FPS</b> , <b>18.4 ms latency</b> and <b>0.854 mAP@0.5</b> on a <b>5.8 W</b> total system power budget |
| Achievements | Co-inventor, <b>Indian Patent App. No. 202641090505</b> (published Sep 2026) · research paper in preparation                              |

### Project bREADth — Non-Invasive Multi-Modal Patient Monitor

Mar 2025 – Present

|              |                                                                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview     | A wearable monitor for continuous vital signs and early collapse detection, with a macOS app for multi-patient floor-plan tracking                          |
| Stack        | Multi-modal sensor fusion, DSP, SwiftUI, MQTT, scikit-learn                                                                                                 |
| Result       | Embedded Random Forest fall detection over <b>104 features</b> , plus a local RAG agentic chatbot for vital-sign diagnostics and clinical handbook querying |
| Achievements | Co-inventor, <b>Indian Patent App. No. 202541059846</b> (published Aug 2026) · manuscript under review at <b>IEEE Sensors Journal</b>                       |

### 8-Channel Thermal Profiling & Heatmap System

Feb 2025

|          |                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview | Eight DS18B20 sensors on a single 1-Wire GPIO, read by an ESP32-S3 acting as a MODBUS RTU slave over RS-485                                            |
| Stack    | ESP32-S3, DS18B20 / 1-Wire, MODBUS RTU over RS-485, Python, PyQt5, scikit-learn, SwiftUI                                                               |
| Result   | A macOS SCADA dashboard with live 2D heatmaps, 3D surface plots, hotspot analysis, CSV logging and <b>real-time Isolation Forest anomaly detection</b> |

### Active Battery Cell Equalization for EV Applications

Sep 2025 – Oct 2025

|          |                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview | An active BMS using a bidirectional flyback converter to move charge between series-connected Li-ion cells instead of dissipating it resistively                                              |
| Stack    | MATLAB, Simulink, power electronics                                                                                                                                                           |
| Result   | A proportional duty-cycle controller with dynamic threshold monitoring drove <b>peak inter-cell divergence from ~1.3 V to near zero</b> , versus persistent imbalance under passive balancing |

## WORK EXPERIENCE

### Systems & Digital Engineering Intern, Boeing India Private Limited

May 2026 – Jul 2026

- Built a fleet maintenance analytics platform (Python, FastAPI, React, TypeScript) that answers queries **across 6,900+ maintenance records in under a second**, with a RAG chatbot on a 5-stage deterministic NLQ pipeline and automated reporting.
- Wrote a forecasting engine that picks among 11 algorithms by horizon over a 24-feature pipeline, cutting error **up to 50%** against linear interpolation.
- Trained an 82-feature XGBoost failure-prediction model (**F1 0.964**, **ROC-AUC 0.952**) with SHAP explainability; its risk alerts save **40+ engineering hours a month**.

### Project Intern, Mindenious Edutech (Remote)

Jul 2025 – Oct 2025

- Completed a VLSI / ASIC design program on commercial EDA tools.
- Wrote Verilog / SystemVerilog RTL with self-checking testbenches, through synthesis to FPGA.
- Measured functional coverage in simulation before taking designs to synthesis.
- Analyzed timing and area trade-offs across synthesis constraints on FPGA targets.

### Project Intern, Apsis Solutions & IIT Guwahati (Remote)

Jul 2025 – Oct 2025

- Built factory-automation monitoring across 13 machines, cutting manual check time **89%**.
- Captured machine status continuously in one live supervisor view.
- Coordinated with IIT Guwahati mentors to validate monitoring accuracy against manual floor logs.

### Summer School, Product Development Management, IIIT Hyderabad

Jul 2024 – Aug 2024

- Studied product lifecycle, market analysis and innovation strategy, then carried one concept from market study to a pitched prototype.
- Presented the prototype and go-to-market plan to a panel of mentors and industry reviewers.

## ACHIEVEMENTS

**1st Place, IITM Gadget Expo** — Best Low-Cost Solution, for Project Garuda  
Co-inventor on **two Indian patent applications** filed for Project Garuda and Project bREADth